FINAL PROJECT REPORT

Unit 4004: Managing a Professional Engineering Project

Predictive Maintenance of Network Infrastructure Using Neural Networks:

A CRM-Integrated Approach for Telecommunications Systems

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Authenticity Declaration

I confirm that this report is entirely my own original work and has not been submitted, in whole or in part, for any other qualification. All sources have been fully acknowledged, and all technical content, analysis and judgements are strictly my own.

Signed:  Date: 2026/10/30

Abstract

This project investigates neural networks for predicting potential telecommunications equipment failures and explores how their outputs can support an alert-handling workflow. A simulated telemetry dataset was used to train a feed-forward baseline and a long short-term memory (LSTM) model. The prototype includes a local API and SQLite mock customer relationship management (CRM) workflow. On 3,576 matched test windows, both models identified all 142 positive windows, but the LSTM generated 3,338 false positives. The saved prediction files support these counts; the model has not been independently re-run for this review. The findings are limited by simulated data, four devices, one test failure event and a time-split boundary issue. Further validation with authorised operational data and a controlled CRM integration is required before operational use.

1. Introduction and Background

Telecommunications infrastructure requires effective maintenance because equipment faults can interrupt services and affect customers. Maintenance may be carried out after a failure, according to a fixed schedule, or in response to equipment condition. This project investigates whether a neural network prototype can use simulated telemetry to identify potential failures before they occur.

A prediction is useful only if its limitations and the actions following it are understood. This project therefore considers both model evaluation and a local alert workflow for human review. Its purpose is to investigate feasibility within a limited prototype, not to establish operational effectiveness or financial savings.

1.1 Aim and Objectives

The aim of this project is to develop and evaluate a neural network prototype for predicting potential telecommunications equipment failures and to demonstrate how its output could be used in a customer relationship management workflow. The four objectives are to: (1) review relevant maintenance and machine-learning literature; (2) prepare telemetry data and document its source and limitations; (3) develop and evaluate a feed-forward baseline and an LSTM using suitable classification metrics; and (4) demonstrate and critically assess a prototype alert workflow, including its ethical and practical limitations.

1.2 Scope

The project is limited to a research prototype using simulated telemetry rather than live telecommunications measurements. The supplied implementation contains a local API and SQLite mock CRM designed to create alerts for human review. It does not include a verified commercial CRM integration, live network monitoring, automated maintenance dispatch or customer communication. Saved prediction outputs were checked against the reported results; the live operation of the implementation was not independently tested during this review.

1.3 Structure of the Report

Section 2 reviews the academic literature underpinning the project's technical choices. Section 3 sets out the methodology and project delivery plan. Section 4 presents the data gathering, feature engineering and modelling work undertaken. Section 5 discusses the results and their implications. Section 6 concludes and recommends future work, followed by a self-evaluation, references and appendices.

2. Literature Review

Predictive maintenance uses equipment data to support decisions about maintenance before a failure occurs. Carvalho et al. (2019) review machine-learning methods for this purpose and show that selecting a method depends on the available data and application. This project applies that general idea to a small simulated telecommunications example; evidence from other industrial settings does not establish performance on network equipment.

A feed-forward model provides a useful baseline for testing whether a more complex sequence model offers an advantage. The LSTM can represent information across an ordered input window, but longer context alone does not guarantee better performance. Both models must be assessed using comparable examples and metrics that reflect missed failures as well as unnecessary alerts.

Data quality and evaluation design matter as much as model choice. Simulated data can support software development when access to operational telemetry is limited, but its patterns may differ from real equipment behaviour. Chronological evaluation helps test predictions on later observations, while testing previously unseen devices would provide additional evidence about generalisation.

Model scores also need a clear route to the people responsible for action. An alert record can provide context for an engineer to review, but a local mock workflow cannot establish the reliability, security or business value of integration with a commercial CRM.

Engineered measurements may help represent changes in equipment condition, but each feature should be treated as a candidate until its contribution is tested. In particular, an experimental thermal-stress proxy must not be interpreted as a validated engineering failure threshold.

The selected approach combines a simple baseline, an LSTM and a prototype alert workflow. Its value depends on the quality of its evidence and on the balance between detection and the workload caused by false alerts.

The comparison below places this approach alongside more familiar maintenance strategies. It is a conceptual comparison, not an experimental benchmark against a threshold-based system.

2.1 Comparison of Maintenance and Monitoring Approaches

Approach	**Description**	**Advantages**	**Limitations**
Reactive	Repair follows a failure.	Simple to organise.	Service may be interrupted before action.
Preventive	Service follows a fixed schedule.	Work can be planned.	Healthy equipment may be serviced unnecessarily.

Approach	Description	Advantages	Limitations
Threshold-based	An alert follows a defined measurement limit.	Straightforward to explain.	A single limit may miss gradual or combined changes.
Predictive	Historical data estimate future failure risk.	Can use multiple measurements over time.	Requires data, validation and management of false alerts.

A neural network can analyse multiple telemetry measurements across time, but it also requires validation and alert management. A future study should compare it with a simple threshold-based method on the same operational data and evaluation criteria (Carvalho et al., 2019).

3. Methodology and Project Delivery Plan

3.1 Methodology

The supplied project files describe an applied machine-learning workflow using synthetic telemetry generated for multiple simulated devices at five-minute intervals. Measurements comprise error rate, latency, temperature and traffic load; derived features include error-rate gradient, latency volatility and a thermal-stress proxy. A feed-forward baseline and an LSTM were evaluated using chronological training, validation and test partitions, with confusion matrices, precision, recall and F1 score. A local API and mock CRM provide a proposed way to handle qualifying predictions.

A positive label indicates a simulated failure within the following 24 hours. The output is a risk score for that prediction horizon, not a prediction of the exact failure time.

3.2 Feasibility

Table 1 summarises the feasibility assessment carried out at proposal stage across four dimensions.

Table 1: Technical, economic, time and resource feasibility assessment

Dimension	Assessment
Technical	Python, TensorFlow and standard data-analysis tools are used in the supplied prototype.
Economic	Open-source tools and free-tier computing may reduce cost; record actual expenditure separately.
Time	The draft plan covers 18 August to 30 November 2026; check meeting frequency and actual progress against the logbook.
Resource	The project requires a computer, suitable telemetry and supervisor access.

3.3 Project Delivery Plan

The project plan sets out a fifteen-week schedule covering research, development, data preparation, evaluation, report writing and supervisor contact. The target dates below describe the plan; actual progress must be supported by the project logbook.

Table 2: Work breakdown structure (see Appendix B for the full Gantt chart)

ID	Task	Details and Deliverables	Duration
1.1	Research & Background	Literature review of AI-based predictive maintenance in telecommunications, IoT/5G fault-prediction studies and neural network architectures; annotated bibliography.	3 weeks
1.2	Development & Analysis	Requirements definition, dataset identification, design of LSTM/feed-forward architecture, environment setup (Python, TensorFlow).	3 weeks
1.3	Data Collection & Analysis	Acquisition and preprocessing of network performance data, feature engineering, model training, hyper-parameter tuning, evaluation using precision, recall and lead time.	5 weeks
1.4	Supervisor Meetings	Progress reviews to discuss methodology and findings.	Ongoing
1.5	Outcome / Report Writing	Development of the working prototype, documentation of results, drafting of the final report.	3 weeks
1.6	Evaluation	Critical evaluation of model performance against objectives, reflection on limitations, recommendations for future work.	1 week

Table 3: Project milestones and target dates; completion status requires confirmation

Milestone	Description	Target Date	Status
Milestone 1	Background research complete	07 Sep 2026	To be confirmed
Milestone 2	Design and environment setup complete	28 Sep 2026	To be confirmed
Milestone 3	Data collection and model evaluation complete	02 Nov 2026	Planned / confirm progress
Milestone 4	Draft report complete	23 Nov 2026	Planned / confirm progress

Milestone	Description	Target Date	Status
Milestone 5	Final submission	30 Nov 2026	Planned / confirm progress

3.4 Risk Assessment

Table 4: Risk assessment and mitigation, as carried into project delivery

Risk	Likelihood	Impact	Mitigation
Suitable telemetry dataset unavailable or incomplete	Medium	High	Document the use of simulated data; obtain authorisation before using operational data.
Model fails to converge or underperforms	Medium	High	Allocate buffer time within the five-week analysis phase; begin with a simpler baseline before an LSTM
Scope creep toward full CRM implementation	Low	Medium	Keep the local prototype clearly distinct from any live commercial CRM integration.
Illness or personal disruption delaying a phase	Low	Medium	Review progress with the supervisor and update the plan when needed.
Data privacy or ethical concerns with network logs	Low	Medium	Review permissions, identifiability and secure handling before using operational records.
Synthetic data may not reflect real networks	High	High	Clearly state the limitation and validate on authorised operational data.
High false-alert volume	Medium	High	Measure precision and false-positive counts; choose a suitable threshold.
Labels cross time-split boundaries	Medium	High	Inspect boundaries and exclude affected examples before reevaluation.
Operational data used without permission	Medium	High	Obtain authorisation and review personal data before use.
Mock CRM mistaken for live integration	Medium	Medium	Label the prototype clearly and verify its capabilities.

The risk assessment identifies delivery, data and model risks. Recorded mitigations should be compared with the project log to distinguish planned actions from actions actually completed.

The additional risks in Table 4 cover the proposed model and alert workflow; mitigations are planned actions unless project evidence shows that they were completed.

3.5 Legal and Ethical Considerations and Their Effect on the Project Plan

Consideration	Relevance	Effect on project plan
Data protection and minimisation	Operational records may identify people or customers.	Obtain authorisation and review and minimise fields before use.
Secure storage	Data and model outputs require protection.	Allow time for access-control and storage checks.
Transparency and human oversight	Risk scores can produce false alerts or missed failures.	Document limitations and provide for an engineer to review alerts.
Representativeness	Performance may differ across devices and conditions.	Plan validation across relevant device groups.

The supplied demonstration telemetry and device identifiers are simulated. Any use of real operational records requires authorisation, an assessment of whether personal data is involved, data minimisation, secure storage and a privacy review. Predictions should be presented as estimates and reviewed by a qualified person before maintenance action.

Sri Lanka's Personal Data Protection Act No. 9 of 2022 was amended by Act No. 22 of 2025. According to the Data Protection Authority, Gazette No. 2498/16 sets 1 January 2027 for commencement of Section 2, Section 3, Part I and Part III. The applicable requirements should be checked at the time of actual data use; no claim of legal compliance is made here (Data Protection Authority of Sri Lanka, 2026; Parliament of Sri Lanka, 2025).

4. Data Gathering and Analysis

4.1 Data Sourcing and Preparation

The saved demonstration uses simulated telecommunications telemetry generated by the project simulator rather than a combination of public and simulated training data. The supplied manifest records 34,560 rows from four devices, sampled at five-minute intervals. Simulated failure events were used to label observations according to whether a failure occurs within the following 24 hours. Simulated data supports workflow testing but limits generalisation to real networks.

The supplied prototype checks required fields, duplicate device timestamps, invalid values and sampling intervals; unsuitable sequential windows are excluded. Its saved data-quality report records no duplicate timestamps or missing required values in the demonstration. Three candidate features are an error-rate gradient, latency volatility and a thermal-stress proxy; their individual contributions have not been established by the supplied evidence.

Table: Dataset summary for the saved demonstration.

Item	Saved demonstration value
Data source	Simulated telemetry
Raw rows / devices	34,560 / 4
Sampling interval	5 minutes
Raw measurements	4
Prediction horizon	24 hours
LSTM input window	144 samples (12 hours)

The system guide mentions a separate fault-record spreadsheet as contextual material rather than a model-training dataset. Its contents and authorisation were not independently checked for this report.

4.2 Model Architecture

The supplied configuration describes two neural-network models. The feed-forward baseline has dense layers of 64 and 32 units with dropout and receives a six-sample input window. The LSTM has layers of 50 and 25 units and receives 144 samples. At five-minute intervals, these windows represent 30 minutes and 12 hours respectively. Both use seven features: error rate, latency, temperature, traffic load, error-rate gradient, latency volatility and an experimental thermal-stress proxy. They produce a binary risk score for the 24-hour horizon.

4.3 Evaluation Design

The supplied code uses a per-device chronological 70:10:20 training, validation and test split, with scaling fitted on training data and a decision threshold selected using validation predictions. Because the model input windows differ in length, their saved test outputs contain different numbers of windows. A fairer comparison matches predictions by device and timestamp. The same simulated devices appear across partitions, so the test examines later periods for known devices rather than performance on unseen devices.

The 24-hour labels were calculated before partitioning. Inspection found validation timestamps whose labels depend on a failure event in the later test partition. This allows information about a test-period event to enter validation and can affect threshold selection. A revised evaluation should exclude boundary-affected samples (or apply an appropriate time gap) and retrain and reevaluate the models; the results in this report have not been rerun with that correction.

4.4 Results

The following values were recalculated from the saved test prediction files using each model's validation-selected threshold of 0.10. The models have different test-window counts, so Table 5 is not a direct like-for-like comparison. Checking saved predictions does not independently reproduce training or inference.

Table 5: Saved test outputs recalculated from prediction files

Measure	Baseline	LSTM
Test windows	4,128	3,576
True negatives	2,465	96
False positives	1,383	3,338
False negatives	15	0
True positives	265	142
Precision	0.161	0.041
Recall	0.946	1.000
F1 score	0.275	0.078

The saved outputs contain 3,576 test windows matched by device and prediction timestamp. The following table uses the same windows for both models, making the comparison more informative, although it remains based on a small simulated sample.

Table 6: Matched-window comparison recalculated from prediction files

Measure	Baseline	LSTM
True negatives	2,189	96
False positives	1,245	3,338
False negatives	0	0
True positives	142	142
Precision	0.102	0.041
Recall	1.000	1.000
F1 score	0.186	0.078

Precision is true positives divided by all predicted positives; recall is true positives divided by all actual positives. The matched counts imply that both models identified the same 142 positive windows, while the LSTM produced substantially more false positives. These are windows, not independent failure events, and high recall on this sample does not establish reliable event-level detection.

The saved metrics contain lead-time calculations for only one test failure event. This is insufficient to establish a reliable average warning time; event-level results should be evaluated over a larger set of failures.

The first submission contained an inconsistency between the baseline precision and its confusion matrix. The tables above use the saved demonstration predictions instead. No model is recommended for live use

on the basis of this evidence. Further work should reproduce the experiment, address split-boundary leakage, assess alert volume at suitable thresholds and validate performance on operational data.

4.5 Prototype and CRM Framework

The supplied implementation includes a local API that accepts telemetry for a device, loads a trained model and returns a failure-risk score. When a score meets the configured threshold, the application is designed to create an alert in a local SQLite mock CRM. The code also provides alert viewing and status updates. Saved files support the design and reported predictions, but the live system was not independently run during this report review.

The mock workflow is a limited prototype, not a commercial CRM integration. It does not establish live monitoring, maintenance dispatch or customer communication. The demonstration should show the model health check, prediction, model-generated alert and review status using the actual system and clearly identified simulated inputs.

Appendix C summarises the alert structure described in the implementation. The actual field names and behaviour should be checked during the demonstration.

5. Discussion

The sections below interpret the results set out in Section 4, assess how the project was delivered against the original plan, and reflect on the professional and ethical dimensions of the work.

5.1 Interpretation of Results

The saved predictions indicate that the models produced risk scores for simulated telemetry, but also show a substantial false-positive burden. On the matched test windows, both models identified the positive windows, while the LSTM generated more false positives than the baseline. High recall alone is insufficient to establish which system is more useful: missed failures and unnecessary inspections both carry operational consequences. The result is limited to this simulation and evaluation procedure and cannot establish performance on live infrastructure.

5.2 Project Delivery

The project plan provides target dates for research, setup, modelling, reporting and submission. The saved model manifest records a demonstration training run on 22 October 2026. Actual milestone completion, changes and supervisor input should be supported by dated records in the project logbook.

The milestones scheduled after 30 October 2026 remain planned at the time of this revision. Project delivery should be updated with actual completion dates when the work is completed.

5.3 Managing People and Stakeholders

Supervisor and stakeholder input should be supported by actual meeting records, participants, feedback and agreed actions. The presentation and professional discussion planned for 21 November 2026 are future activities as of 30 October 2026; their outcomes cannot yet be reported as completed.

5.4 Professional Standards and Reflection

Professional practice requires transparent reporting of the evidence and its limitations. The supplied telemetry is simulated; a separate fault-record spreadsheet may contain operational identifiers and requires authorisation and a privacy assessment before use. A false positive can cause unnecessary work, while a false negative can leave a developing fault undetected. Alerts should be reviewed by an appropriately qualified person, and the mock system should not be presented as production-ready.

5.5 Critical Evaluation of Project Outcomes

The supplied files describe a workflow connecting simulated telemetry, neural-network predictions and a local mock alert system. The saved test predictions support the reported counts, although a live demonstration was not independently rerun during this review. A small simulated dataset cannot establish readiness for use in a live telecommunications network.

Area	Strength	Limitation
Model development	Two architectures are available for comparison.	Saved outputs were matched; inference was not independently rerun.
Data preparation	The described workflow checks telemetry and creates features.	Simulated data may not reflect network behaviour.
Evaluation	Saved predictions support the reported classification measures.	Few devices and failure events limit confidence.
CRM workflow	A local mock can illustrate alert handling.	Commercial integration and security are untested.
Explainability	Prediction and model version can provide context if recorded.	Individual feature explanations are not established.
Practical application	The design illustrates a possible review process.	Costs, downtime and customer impact were not measured.

On the reported matched windows, the LSTM's high recall accompanies low precision and many false positives. If reproduced in practice, this would create substantial review work and could reduce trust in alerts. The evidence also does not establish how either model performs on unseen devices, different network conditions or a larger set of failure events. The local mock demonstrates an intended workflow, but does not test commercial integration, security or customer communications.

The next steps follow from these limitations: obtain authorised and representative telemetry, test more devices and failure events, check time-split boundaries, compare matched predictions, select thresholds against operational costs, and measure alert workload and response time during a controlled pilot.

6. Conclusions and Recommendations

This project investigated neural networks for predicting potential telecommunications equipment failures using simulated telemetry. The supplied system includes a feed-forward baseline, an LSTM and a local mock CRM intended to illustrate alert handling. Counts recalculated from the saved prediction files agree with the results presented in this report; the system was not independently run for this review.

The saved results demonstrate that the recorded workflow produced predictions and classification metrics, but they also indicate a high false-positive burden and limited failure-event evidence. Together with the split-boundary issue, these limitations prevent a conclusion that the system is ready for operational deployment.

The prototype provides a basis for future validation using authorised real-world telemetry and a controlled CRM integration. Its practical value remains unmeasured: this project does not establish reductions in downtime, maintenance expenditure or customer impact.

Future work should validate on representative authorised telemetry; include more devices and failure events; compare models on matching test windows; check split-boundary leakage and unseen-device performance; select an alert threshold with the maintenance team; test a controlled CRM sandbox; and measure alert workload and response time before considering deployment.

7. Self-Evaluation

The following reflection draws on the project account and should be read alongside the project logbook.

Section One: The experience/task being reflected on

This project required me to plan and evaluate a predictive-maintenance prototype for telecommunications equipment and consider how its predictions could be reviewed in a CRM workflow. It brought technical work together with project planning, risk assessment and communication. My role involved following the progress from an initial idea through data preparation, model development, evaluation and consideration of an alert workflow. The available system files describe a simulated dataset, two trained model architectures and a local mock CRM. The project plan sets out the stages and target dates, while the report records the results and their limitations. In this reflection I focus on how I approached the technical choices, what the comparison revealed, and how I would improve the evaluation. I also consider what the

project taught me about being cautious when a result looks encouraging but the underlying evidence is limited.

Section Two: Steps taken and why

I began with a review of predictive maintenance and considered how equipment measurements might be used to anticipate failures. Since the demonstration used simulated telemetry, I treated the data as a way to test the workflow rather than as proof of real-network performance. The dataset contained error rate, latency, temperature and traffic load, with additional derived features for trends and possible stress. The modelling approach used a feed-forward network as a baseline and an LSTM to examine a longer sequence. This made it possible to compare the more complex model with a simpler starting point. The evaluation used chronological partitions and examined precision, recall and F1 rather than accuracy alone. I also considered how a prediction might be routed through a local API into a mock CRM, where a person could review the alert. This workflow connected the technical result to an action while keeping the scope limited to a prototype.

Section Three: What went well

Comparing two models was useful because it exposed differences that a single score would have hidden. The matched-window comparison used the same device timestamps for each model. Both identified the positive windows in that comparison, but the LSTM produced substantially more false positives. This showed why a high recall value needs to be considered alongside precision and the amount of work generated for reviewers. I found the distinction between prediction windows and independent failure events particularly useful when interpreting the results. The dataset also made it possible to exercise several parts of a coherent prototype, from telemetry preparation to a proposed alert record. The saved outputs, configuration and data-quality report provide a basis for reviewing the calculations. These records are valuable because another reader can follow which data and model settings were associated with the reported demonstration, even though an independent rerun remains necessary.

Section Four: What could have been done better

The project would have benefited from a clearer record connecting every result to its input data, model version, evaluation split and decision threshold from the outset. An inconsistency in the earlier baseline report illustrates why numerical claims should be recalculated from the underlying predictions before submission. The split-boundary issue also showed that a chronological split alone is not enough when labels use a future failure horizon: a validation label can depend on an event in the later test period. Boundary-affected samples need to be excluded and the models evaluated again. The simulation has only four devices and the saved event-level calculation covers one test failure, so it cannot provide confidence in generalisation. I would also set aside more time to understand alert workload and to test the workflow in

conditions closer to its intended use. A stronger project record would distinguish planned actions from completed work and connect each decision with supporting evidence.

Section Five: What has changed or improved as a result

This work reinforced the importance of checking reported metrics against predictions and assessing missed failures alongside false alarms. In future work I would plan boundary checks before training, retain reproducible experiment records and compare models on matched examples. I would also include tests across previously unseen devices and more failure events where suitable data are available. The mock CRM demonstrates why a useful prediction needs a clear route to human review, but it does not establish the security or performance of a commercial integration. Before considering operational use, I would seek authorised data, agree an alert threshold with the people who would review the alerts, and measure the practical workload created by them. For project management, I would prepare specific questions for supervisor meetings and record decisions and actions against the plan. These steps would make the reasoning easier to communicate and allow later changes to be assessed against the original objectives. I would also document the assumptions behind each demonstration so that limitations are clear to the audience reviewing the work.

8. References

Carvalho, T.P. et al. (2019) 'A systematic literature review of machine learning methods applied to predictive maintenance', Computers & Industrial Engineering, 137, 106024. Available at: https://doi.org/10.1016/j.cie.2019.106024 (Accessed: 29 October 2026).

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Parliament of Sri Lanka (2025) Personal Data Protection (Amendment) Act, No. 22 of 2025. Available at: https://www.parliament.lk/uploads/acts/gbills/english/6384.pdf (Accessed: 29 October 2026).

Appendices

Appendix A: Project Proposal Form

The project proposal, feasibility study and project plan provide the planned scope and delivery approach. They should be supplied separately where required by the assessment.

Appendix B: Project Gantt Chart

The project Gantt chart records the planned schedule. The accompanying project records should distinguish target dates from actual progress.

Appendix C: CRM Alert Record Draft Field Structure

The following fields summarise the intended alert record. The prototype stores a risk score, device identifier, generation time, priority, model version and review status; the decision threshold is returned with the prediction.

Field	Purpose in the prototype
Device identifier	Identifies the simulated device associated with an alert.
Risk score	Records the model output for review.
Threshold	Records or identifies the configured alert threshold.
Timestamp	Records when the alert was created.
Review status	Records the status of human review.
Model version	Identifies the model used, if recorded.

Appendix D: Project Supervisor Meetings Log

The project logbook and supervisor meeting records provide the dates, topics, feedback and actions agreed during project delivery.

Appendix E: Presentation

The project presentation is a separate supporting item. Any delivery or professional discussion evidence should reflect the actual event and date.

Appendix F: Supplementary Portfolio Evidence

The supplementary evidence comprises the project system, simulated data, model configuration, saved predictions and evaluation results. Presentation and project-management evidence should be provided separately as required.